

Kletetschka's Three-Dimensional Time and Blumberg's QGTCD-SIT Genealogy

An atomized four-lens priority comparison with versioned prior-art and longitudinal source control

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Abstract

A falling object changes its average motion toward mass. A theory of gravity must explain what physical state differs from place to place, how that difference changes motion, and how the local operation scales into the observed gravitational trajectory. In June 2022, Micah Blumberg publicly described a proposed answer: mass and oscillation alter local temporal structure; a gradient in temporal opportunity makes otherwise directionally indifferent particle motion more probable toward mass; repeated local biases produce gravitational attraction. The name **Quantum Gradient Time Crystal Dilation (QGTCD)** was applied later to this proposed operation. Gunther Kletetschka's three-dimensional-time program entered the checked record with a manuscript received on November 27, 2024 and published on April 21, 2025. It uses different mathematics: three orthogonal temporal coordinates, a six-coordinate metric, temporal eigenvalue equations, and reported particle and cosmological predictions. Yet it reaches a bounded cluster of matching conclusions: temporal structure is physically fundamental; mass and matter manifest temporal structure; temporal dynamics generate gravitational behavior; temporal organization spans quantum, interaction, and cosmological scales; and time supplies a route toward quantum-gravity unification. A September 2025 paper then explicitly makes familiar three-dimensional space a projection from an underlying temporal manifold.

This paper does not reduce the comparison to identical equations or identical terminology. It decomposes the programs into inputs, transformations, outputs, constraints, and dated atoms. A full-genealogy saturation pass recovers three additional earlier partials that a late-terminology search had missed. Ten target-relative atoms receive bounded **Micah-earlier partial-priority** dispositions. The strongest shared directed sequence is: structured temporal substrate → temporally varying state → matter / mass and gravitational behavior → quantum-gravity unification, with later Kletetschka work adding explicit space projection. Kletetschka retains credit for the exact orthogonal metric, temporal eigenvalue spectrum, reported numerical predictions, charge-as-temporal- winding proposal, and projection functions. Bars's 1998 two-time physics and Chen's 2005 3+3 model are genuine earlier prior art for broad multi-time atoms. They do not erase Blumberg's narrower time-density-gradient and stochastic-directional-bias delta, nor do they contain the later time-primary, matter-as-temporal-manifestation, and space-from-time projection cluster in the checked versions.

Four independent comparison lenses are applied: bounded mechanism equivalence, directed narrative architecture, conclusion-first reconstruction, and longitudinal signature migration. The first lens passes partially; the second and third pass; the fourth finds a bounded component shift and persistence across Kletetschka's April, June, and September 2025 physics papers and February 2026 presentation. A mandatory newest-source check also identifies a June 9, 2026 Kletetschka-Walker preprint that applies three orthogonal temporal axes to intracerebral, interpersonal, and evolutionary information transmission. Because that source changes the coauthor, target domain, equations, and outputs, it is treated as an adjacent joint program and a source-gated SAN comparison - not as an eleventh certified physics atom. The probability status remains **quantitative-not-calibrated**: descriptive atom counts establish content and chronology but do not establish five-sigma rarity, access, copying, dependence, or intent.

Theory; time-density gravity; temporal geometry; priority; causal-interface sequence; longitudinal signature migration; multiscale information transmission; human evolution; theory comparison

Keywords: three-dimensional time; QGTCD; Super Dark Time; Super Information

Scope and evidence boundary

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1. The problem before the terminology

The ordinary observation is simple: release an object near Earth and its average path changes toward Earth. General relativity explains the observation through spacetime geometry. A proposed replacement or extension must do more than rename gravity. It must identify a physical state, specify how that state varies, give a rule connecting the variation to motion, recover known limits, and expose a measurement capable of rejecting the proposal.

Comparing two unconventional theories creates a second problem. Different authors can use different objects, equations, examples, and metaphors while preserving the same bounded causal operation. Conversely, two theories can use the same word - *time*, *density*, *curvature*, *information*, or *dimension* - while performing different operations. A word-match is therefore too weak. Requiring the entire theories to be identical is too strong.

The workable unit lies between those extremes. For each claim, ask:

1. What enters the operation?
2. What changes?
3. What constraint selects the change?
4. What leaves the operation?
5. Which part of the sequence is publicly dated?
6. Which parts are shared, earlier, later, or genuinely different?

Only after this plain operation is visible does this paper introduce its compact method labels. A **bounded atom** is one source-located operation, causal arrow, output, or constraint. A **causal-interface sequence** is an ordered chain of those roles rather than an unordered list of similar nouns. The four lenses used below test mechanism, directed narrative, conclusion-first reconstruction, and program change over time.

The central question is not whether QGTCD and Kletetschka's theory are the same full theory. They are not. The question is whether Kletetschka later reaches bounded conclusions and causal roles that Blumberg publicly articulated earlier, even though Kletetschka implements them with different ontology and mathematical machinery.

2. Claim contract

This paper keeps five statement types separate.

1. **Documentary statements** report what a fixed source says and when the checked source became public or entered a journal workflow.
2. **Priority statements** identify the earliest verified public source among the compared formulations for one bounded atom.
3. **Equivalence statements** identify a preserved operation, composition, or observable trace under an explicit translation.

4. **Scientific statements** distinguish a model’s internal derivation or reported prediction from independent physical reproduction.
5. **Historical-causal statements** would concern access, transmission, dependence, copying, or intent. This paper makes none of those statements.

The priority result is affirmative but bounded. A later theory can be mathematically more explicit and still arrive later at a causal conclusion. An earlier source can own a generic class without owning every later delta. One earlier atom cannot create an author-wide floor, and one unmatched residual cannot erase every shared atom. This is the paper’s **patent-atom discipline**: credit and difference are allocated row by row. The discipline is used as a content-architecture method; no legal infringement finding is made.

3. Source and date method

3.1 Date types are not interchangeable

An immutable public Git commit establishes that text was fixed in a public repository by that commit. An arXiv v1 date establishes public preprint availability. A journal receipt date documents editorial intake but does not necessarily establish public availability. Acceptance and online publication are different events. A repository metadata field supplied by an author is not automatically an independent archive capture. The chronology below preserves those distinctions.

3.2 Version identity matters

The same arXiv identifier may contain materially different versions. Chen’s 2005 v1 is 14 pages; the August 2024 v2 is a 47-page rewrite. V1 already contains a 3+3 multi-time model, an extra-time account of quantum phenomena, a causality discussion, a Kaluza-Klein gravity / electromagnetism unification program, and a compactification mass spectrum. V2 adds a strong-interaction equation and a larger gravity / electromagnetism / strong-interaction unification claim. Those additions cannot be backdated to 2005.

The same rule applies to Blumberg. A 2026 normalized transition kernel can clarify the 2022 prose, but it cannot become a 2022 equation. Later SDT and SIT actions can extend the genealogy without replacing or backdating the first QGTCD disclosure.

3.3 Search stance and adversarial check

The discovery pass begins as a priority-supporting search: it looks for the strongest dated Blumberg formulation and translates target vocabulary rather than waiting for identical words. The certification pass then tries to defeat that case through earlier sources, version changes, generic field trends, target-specific residuals, missing links, and falsification conditions. This sequence is advocacy followed by adversarial certification, not neutrality through premature omission.

3.4 Full-genealogy saturation before an adverse disposition

A chronology page or one late technical manuscript is a source router, not proof that Micah’s earlier source family was searched. Before a target atom can be called absent, target-specific, target-earlier, or content-insufficient, the audit must translate that atom into the ordinary vocabulary used at every applicable Micah stage, search the bounded source families, and materialize the stage-by-target-atom result. An unsearched cell remains source-gated; it is not negative evidence.

That rule changes this revision. Draft 6 searched historical QGTCD mainly through the later gravity terminology and missed three earlier formulations:

1. a June 2022 source explicitly asking whether time has three dimensions;
2. a 2023 quantitative proposal relating mass, frequency, and local time-frame number; and
3. a June 2022 charge model using oscillatory orientation, phase relation, and temporal interval.

The later SVGN comparison article is also preserved as a dated public synthesis [26]. It describes three-dimensionality as implicit in 2022 and explicit in 2025 SIT. The immutable b0020y object is stronger on this point because its 2022 wording already says “three time dimensions” [22]. The later article therefore strengthens the public comparison record without weakening or replacing the earlier source.

4. Typed chronology

Date	Date type	Source	What the event establishes
1998-09-04	arXiv v1 submission	Bars, <i>Two-Time Physics</i> [5]	Earlier two-time physics using $SO(d, 2)$ and $Sp(2)$ structure.
2005-10-03	arXiv v1 submission	Chen, <i>3-Dimensional time</i> , v1 [6]	Earlier 3+3 time, quantum and causality treatment, gravity / electromagnetism unification, and compactification mass spectrum.
2022-06-08	public Git commit	Blumberg, a0010z_ctpr.txt [8]	Oscillators generate or define spacetime; mass may be denser / slower spacetime; temporal scale depends on oscillator, mass, density, and scale.
2022-06-08	public Git commit	Blumberg, b0020y_ctpr.txt [22]	Explicitly asks whether time has three dimensions and names length-time, width-time, and height-time.
2022-06-08	public Git commit	Blumberg, original a0033z.md [23]	Charge polarity is modeled through perpendicular oscillatory orientation and splay relation; particle identity is linked to time between oscillation intervals.
2022-06-16	public Git commit	Blumberg, a0033z.md [9]	Explicit “gravity is gradient time dilation” statement plus otherwise indifferent motion biased toward mass by increased temporal opportunity.
2022-06-30	public Git commit	Blumberg, refined b0020y.md [28]	Restates the need to describe time in 3D, joins three spatial and three temporal dimensions, and relates them to particle trajectories and gravity.
2022-07-26	public Git commit	Blumberg, a0253z.md [10]	Mass / time-crystal, local frequency / density change, time change, gravity, and motion toward mass joined in one note.
2022-07-30	public Git commit	Blumberg, a0253z.md [11]	Directional-probability account restated using additional temporal measure.

Date	Date type	Source	What the event establishes
2022-11-28	public Git genealogy	QGTCD name fixation [12]	The later name is attached to the already developing mechanism.
2023-09-15	public Git commit	Blumberg, public transcript in y0021.md [24]	States that particle mass, frequency, and local time-frame number are interconnected and that the time-frame factor modifies wave-function timing.
2023-12-10	public Git commit	Blumberg, b0102ySci.md [27]	Refines the charge hypothesis through different time windows, time-frame quantities, and alternating-frequency pairing.
2023-12-15	public Git commit	Blumberg, 00change1og.md [25]	Preserves speculative mass-frequency-time-frame equations and explicitly labels their evidentiary status.
2024-01-29	public article date	QGTCD Part I [13]	Public unified-field presentation of the time-density / trajectory proposal.
2024-02-15	public article date	QGTCD core article [14]	Probabilistic geometry and gravity beyond metric-curvature-only language.
2024-03-28	public article date	QGTCD plain-language article [15]	Biased-random-walk explanation of changed directional odds.
2024-04-21	independent archive capture	QGTCD core and plain-language pages [16,17]	Independent Wayback evidence that both public routes existed by this date.
2024-08-20	arXiv v2 submission	Chen v2 [7]	Expanded field theory and strong-interaction material; this is a 2024 event, not a 2005 event.
2024-11-27	journal receipt	Kletetschka April paper [1]	The journal received the target manuscript on this date.
2025-01-29	author-supplied record metadata	SDT v16 [18]	Exact current public artifact used for later genealogy and equation-role comparison; not independent proof of exact-PDF availability on that date.
2025-02-09	author-supplied record metadata	SIT v79 [19]	Exact current public artifact used for later genealogy and equation-role comparison; not independent proof of exact-PDF availability on that date.
2025-02-24	journal acceptance	Kletetschka April paper [1]	Acceptance event.
2025-04-21	journal online publication	Kletetschka April paper [1]	Public paper containing the explicit 3+3 metric, time-primary narrative, and reported predictions.
2025-06-09	journal online publication	Kletetschka charge paper [2]	Adds charge as temporal winding and possible $U(1)$ emergence.

Date	Date type	Source	What the event establishes
2025-07-25	public article date	Blumberg SVGN comparison [26]	Publicly assembles the Kletetschka / QGTCD-SDT-SIT priority argument and routes a wider field of similar theories; it is a later synthesis, not the date of the older atoms it cites.
2025-09-30	journal online publication	Kletetschka space-projection paper [3]	Explicitly makes familiar three-dimensional space a projection from a temporal manifold.
2026-02-06	public university presentation	Kletetschka UAF seminar [4]	Newest checked physics-program source preserves projection and coupled scale-dependent temporal axes.
2026-06-09	SSRN posting	Kletetschka and Walker [21]	An adjacent joint program applies three orthogonal temporal axes to intracerebral processing, interpersonal learning, and cumulative cultural / evolutionary transmission.

This chronology proves neither physical correctness nor access. It does establish three facts relevant to priority:

1. generic multi-time and 3+3 proposals precede both Blumberg and Kletetschka;
2. Blumberg's public time-density-gradient and directional-probability gravity operator precedes the checked Kletetschka program;
3. Blumberg also publicly fixed an explicit three-dimensional-time concept and a temporal / phase account of charge in 2022, followed by a quantitative mass-frequency-local-time relation in 2023;
4. Kletetschka's later physics program expands toward exact orthogonal-metric, eigenvalue-spectrum, winding, and space-projection implementations after those bounded Blumberg disclosures; and
5. the June 2026 Kletetschka-Walker source extends the three-axis formalism into predictive neural, interpersonal, and evolutionary information processing, but remains a distinct joint program whose strict SAN-priority disposition is source-gated.

5. The earlier multi-time floor, atom by atom

5.1 Bars 1998

Bars develops a two-time framework in a $d+2$ setting. Its characteristic structure is an $SO(d, 2)$ symmetry and $Sp(2)$ gauge relation among one-time systems [5]. This is real earlier prior art for the broad proposition that more than one temporal dimension can participate in fundamental physics.

It does not, in the checked paper, supply:

- a local scalar time-density gradient;
- a stochastic directional-bias rule toward mass;
- exactly three orthogonal temporal axes;
- matter as a property of time;

- an explicit space projection from a temporal manifold; or
- Kletetschka's reported temporal eigenvalue spectrum.

Bars therefore narrows broad originality language without deciding the present delta.

5.2 Chen 2005 v1

Chen v1 is more directly relevant. It treats three spatial and three temporal dimensions, interprets quantum behavior through two additional temporal dimensions, addresses common causality objections, places gravity and electromagnetism into six-dimensional Kaluza-Klein equations, and derives a compactification-based mass spectrum [6]. Kletetschka cites Chen and explicitly distinguishes his physical implications [1].

These are genuine Chen-earlier atoms. They prevent claims that either Blumberg or Kletetschka invented 3+3 time, multi-time quantum interpretation, generic multi-time causality, or six-dimensional gravity / electromagnetism unification.

But v1 does not state the narrower QGTCD operation:

$$\begin{aligned}
 &\text{local time-density difference} \\
 &\rightarrow \text{unequal directional opportunity} \\
 &\rightarrow \text{mean motion toward mass.}
 \end{aligned} \tag{1}$$

Nor does it state Kletetschka's later narrative that time is the primary fabric, matter is a property of time, and familiar space is projected from time.

5.3 Chen 2024 v2

Chen v2 preserves the arXiv identity but is a different scientific stage [7]. It expands the equations and adds a strong-interaction derivation and a claim to unify gravitational, electromagnetic, and strong interactions in one six-dimensional Kaluza-Klein equation. This source predates Kletetschka's manuscript receipt, so it matters to the target's broad unification context. It does not move those added atoms back to 2005.

The resulting allocation is not "Chen owns the author." It is:

- Chen earlier on the 3+3 class and named v1 atoms;
- Chen v2 earlier than Kletetschka on its added 2024 atoms;
- Blumberg earlier on the time-density-gradient stochastic-gravity delta;
- Kletetschka later but distinct on its exact metric, scale-axis assignments, eigenvalue machinery, predictions, charge topology, and projection functions.

6. Blumberg's public mechanism genealogy

6.1 Ancestral spacetime-from-oscillation structure

The June 8, 2022 note does not yet provide a finished gravity theory. It does provide the conceptual spine from which the later operator develops. Oscillators are described as defining or creating spacetime. Mass is considered as slow-moving denser spacetime. The apparent scale of time is linked to oscillator, mass, density, and physical scale. Quantum and macroscopic regimes are therefore not merely different locations in one passive time parameter; they are regimes whose temporal behavior depends on physical organization [8].

That note supplies four later-relevant atoms:

1. temporal structure is physically produced, not merely a coordinate label;
2. mass is tied to concentrated temporal / spacetime structure;
3. space and time arise together from oscillatory relations; and
4. temporal organization is scale-dependent.

6.2 Gradient, coincidence, and directional bias

On June 16, 2022 the causal operation becomes explicit. The note states that gravity is gradient time dilation at particle and atomic scale. It then asks what happens when a particle has no preferred direction in an approximately uniform environment but has more temporal opportunities in the direction of a large mass. The answer is not that the particle consciously selects mass. The local transition ensemble is no longer directionally symmetric, so repeated steps have a biased mean [9].

The historical shorthand can be written

$$\mathbf{g} \propto \nabla \rho_t, \quad (2)$$

where ρ_t denotes the proposed local temporal measure or density. Equation (2) is a compact directional hypothesis, not a dimensionally complete law. A testable phenomenological form must introduce a response coefficient,

$$\mathbf{g} = \gamma \nabla \rho_t, \quad (3)$$

define the units and dynamics of ρ_t , derive or measure γ , and explain why free-fall acceleration is composition-independent.

The July notes join this operator to mass-as-time-crystal language and make the directional-probability interpretation more explicit [10,11]. The 2024 public articles then explain the same chain through unequal local time measure and a biased random walk [13-17].

6.3 Recovered three-time, mass-frequency, and charge stages

The full-genealogy pass adds three earlier deltas that cannot be recovered by searching only for the later target's exact vocabulary.

First, the June 8, 2022 b0020y source asks whether time itself requires three dimensions and names length-time, width-time, and height-time [22]. The June 30 refinement says that time may need to be described in 3D, places three temporal dimensions beside three spatial dimensions, and links that possibility to particle trajectories and gravity [28]. This is earlier priority for the explicit three-dimensional-time concept. It is not Kletetschka's later orthogonal metric, field equation, or numerical implementation.

Second, the September 2023 public transcript joins particle mass, frequency, and local time-frame number and proposes that the time-frame factor changes wave-function timing [24]. The December 2023 public changelog preserves the speculative relations $mc^2 = (\hbar/N) f$, $E = (\hbar/N) f$, and $m = \hbar / (Nf)$, together with a position- and time-dependent $N(\mathbf{r}, t)$ proposal [25]. The source itself discloses the equations as speculative and AI-assisted. That disclosure limits validation claims; it does not erase the dated quantitative relation. Kletetschka retains the exact temporal eigenproblem, exponential generation spectrum, and reported numerical outputs.

Third, the original June 2022 a0033z source explains positive and negative charge through different oscillatory orientations, phase / splay relations, and temporal intervals [23]. A December 2023 refinement

uses different time windows, time-frame quantities, and alternating-frequency pairing to explain attraction and repulsion [27]. This is an earlier temporal / phase architecture for charge. It is not an integer winding invariant, a derivation of $U(1)$, or charge quantization. Those exact later implementations remain Kletetschka's.

These are partial-priority findings because the shared functional deltas precede the target while the exact target operators remain residuals. Requiring the entire later operator before recognizing an earlier partial would repeat the full-operator-or-zero fallacy.

6.4 A later mathematical target, not a backdated equation

If $X_n = x$ is a current state, $y \sim x$ is a neighboring state, and $\rho_t(x) > 0$, one possible later formalization is

$$\Pr(X_{n+1} = y \mid X_n = x) = \frac{W(x, y; \rho_t)}{\sum_{z \sim x} W(x, z; \rho_t)}. \quad (4)$$

Equation (4) exposes what the historical prose leaves open. The weighting function W must be selected, normalized, related to inertial dynamics, and constrained by relativity. It is not a historical 2022 QGTCD equation.

Only after the problem, operation, and open mathematics are clear is the name useful: **Quantum Gradient Time Crystal Dilation** names the proposal that mass-associated oscillatory structure changes local temporal measure, its gradient changes transition odds, and the accumulated bias appears as gravity.

6.5 Later SDT and SIT stages

Super Dark Time and Super Information Theory extend the same genealogy toward scalar-field actions, coherence, information, thermodynamics, dark-sector claims, and broader unification [18,19]. In the exact current public artifacts, a scalar time-density field lives on ordinary Lorentzian spacetime. SIT adds a complex coherence field and couplings to the time-density sector.

These later artifacts strengthen the persistence and scope of Blumberg's time-centered program. They do not convert a scalar field into three orthogonal temporal coordinates, and they are not used to backdate their equations into 2022.

7. Kletetschka's evolving program

7.1 April 2025: metric, time primacy, and reported predictions

Kletetschka begins from a six-coordinate interval with three temporal and three spatial coordinates [1]:

$$ds^2 = dt_1^2 + dt_2^2 + dt_3^2 - dx^2 - dy^2 - dz^2. \quad (5)$$

The field-equation form is reported as

$$R_{ab}(T, x) = \kappa T_{ab}(T, x), \quad (6)$$

and the multi-temporal wavefunction uses frequency components across t_1, t_2, t_3 . The propagator is reported in the form

$$G(k, \omega) = \left[k^2 + \omega_1^2 + \omega_2^2 + \omega_3^2 \right]^{-1}. \quad (7)$$

Temporal dependence is then used in an eigenvalue relation,

$$\left(\partial_{t_1}^2 + \partial_{t_2}^2 + \partial_{t_3}^2 \right) \Psi_n = -m_n^2 \Psi_n, \quad (8)$$

with an exponential mass relation of the reported form

$$m_n = m_0 \exp(-\alpha n \gamma), \quad n = 1, 2, 3. \quad (9)$$

The paper reports a generation ratio 1 : 4.5 : 21.0, particle and neutrino masses, weak-mixing and CP quantities, new resonance scales, gravitational-wave effects, and cosmological signatures [1]. This comparison records them as **reported-not-independently-reproduced**. A source can report a close numerical match without this paper having reconstructed the parameter selection, boundary conditions, uncertainties, or sensitivity.

The narrative content is as important to the comparison as the metric. The primary paper states that time might be the primary fabric of the universe, that mass and energy are manifestations of temporal structure, that matter is a property of time itself, and that three temporal scales generate quantum, interaction, and cosmological phenomena [1]. These are not merely press-release phrases; they appear in the primary paper.

7.2 June 2025: charge as temporal topology

The second paper proposes that electric charge is a temporal winding number and that charge quantization and conservation may arise from topology in the temporal manifold. It distinguishes mass, tied to temporal eigenvalues, from charge, tied to winding. It also proposes that U(1) symmetry may follow from temporal rotational invariance [2].

No checked Blumberg source supplies this exact charge-as-winding operator. It remains a Kletetschka-specific addition in this comparison.

7.3 September 2025: space becomes an explicit projection

The third paper changes the compared architecture materially. It proposes a basic one-spatial plus three-temporal structure and says that familiar three-dimensional space evolves as a projection from the underlying temporal manifold. It reports mapping functions between the basic and apparent dimensions and joins electromagnetic and gravitational interactions through temporal curvature [3].

The April paper already makes time primary and matter a manifestation of time. The September paper adds the explicit space-from-time transform. This is not merely the same sentence repeated. It is a later program stage that moves closer to the time-first, generated-space architecture already present in Blumberg's 2022 genealogy.

7.4 February 2026: newest checked physics-program source

The newest checked source in the fundamental-physics family is a University of Alaska seminar abstract. It again makes familiar three-dimensional space an effective projection of the temporal manifold and describes microscopic, macroscopic, and cosmological regimes as governed by distinct but coupled temporal axes [4]. The projection and multiscale atoms therefore persist; they are not a one-paper accident.

7.5 June 2026: adjacent cognition-and-evolution program with Walker

The mandatory newest-source search recovers a later, separately authored target: Kletetschka and Michael J. Walker’s 39-page SSRN preprint *Multiscale Information Transmission in Human Evolution: Towards a Three-Dimensional Temporal Formalism*, posted June 9, 2026 [21]. Its target object is not particle physics. It represents information transmission across three coupled scales:

1. intracerebral neural processing over milliseconds;
2. interpersonal learning over individual lifetimes; and
3. cumulative cultural and evolutionary transmission over much longer intervals.

The abstract reports three orthogonal temporal axes for those scales, explicit coupling terms, projection onto ordinary one-axis clock time, predictive-memory change as a vector in the three-dimensional temporal space, and a temporal integrated-information measure. It connects the construction to active inference, the Zone of Bounded Surprisal, the Baldwin effect, working-memory expansion, white-matter maturation, and the archaeological record [21].

This is evidence of a new adjacent program stage, but not permission to merge unlike objects. The coauthor, domain, source set, equations, outputs, and falsification targets all change. The present paper therefore records two separate propositions:

- the Kletetschka physics program contains ten certified target-relative Micah-earlier partial atoms; and
- the Kletetschka-Walker cognition / evolution program has a positive bounded architecture match to earlier SAN genealogy, while strict priority remains source-gated until the full primary text, exact locators, named-prior search, and stage-by-target-atom saturation matrix are complete.

That separation prevents two opposite errors: omitting the newest target because it belongs to another domain, or inflating the established physics count by silently treating a joint cognition paper as the same program.

8. Atom-by-atom priority canvas

The table reports bounded relations, not whole-author verdicts.

Atom	Earliest checked Micah formulation	Checked Kletetschka formulation	Relation and priority
A01 Temporal primacy	2022-06-08: oscillators generate or define spacetime [8]	2025-04-21: time may be primary fabric / fundamental substance [1]	Strong narrative partial mapping; Micah earlier
A02 Time-density or temporal-medium role	2022-06-08: mass, frequency, density, and scale alter temporal structure [8]	2025-04-21: temporal curvature / dynamics structure physical behavior [1]	Medium operator-role mapping; Micah earlier
A03 Matter / mass from temporal structure	2022-06-08: mass as slower, denser spacetime [8]	2025-04-21: mass / energy manifestations of temporal structure; matter a property of time [1]	Strong conclusion mapping; Micah earlier
A04 Gravity from temporal difference	2022-06-16: gradient time dilation produces directional attraction [9]	2025-04-21: temporal geometry / curvature produces gravitational behavior [1]	Strong bounded mechanism / narrative mapping; Micah earlier
A05 Quantum-gravity temporal unification	2022-06-16: particle-scale probability process proposed as gravity [9]	2025-04-21: three time axes proposed to unify quantum and gravitational phenomena [1]	Strong causal-scope mapping; Micah earlier

Atom	Earliest checked Micah formulation	Checked Kletetschka formulation	Relation and priority
A06 Space derivative from time	2022-06-08: oscillators generate relative space and time together [8]	2025-09-30: perceived three-space projected from temporal manifold [3]	Medium architecture mapping; Micah earlier on direction, Kletetschka owns projection implementation
A07 Multiscale temporal organization	2022-06-08: time scale depends on oscillator, mass, density, and scale [8]	2025-04-21 and 2026-02-06: quantum, interaction, and cosmological axes [1,4]	Strong role mapping; Micah earlier
A08 Stochastic directional bias	2022-06-16: more temporal opportunity changes otherwise indifferent directional odds [9]	Not recovered in checked Kletetschka sources	Micah-only residual
A09 Coherence / information field	Later SIT stage [19]	Not recovered in checked Kletetschka sources	Micah-only residual; not backdated
A10 Multi-time causality	Later checked Micah artifact [18]	2025 paper claims causality and unitarity [1]	Source-gated: Chen 2005 earlier generic precursor; exact Micah-side well-posedness / causality operator not recovered
A11 Three-dimensional time / exact orthogonal metric	2022-06-08: explicit three time dimensions, including length-, width-, and height-time [22]	Exact orthogonal 3+3 metric and Equation (5) [1]	Micah earlier on explicit 3D-time concept; Kletetschka owns exact metric implementation; Chen earlier on generic mathematical 3+3
A12 Mass-frequency-local-time relation / exact eigenvalue spectrum	2023-09-15 and 2023-12-15: mass, frequency, and local time-frame number joined quantitatively [24,25]	Temporal eigenvalue Equations (8)-(9), generation spectrum, and reported values [1]	Micah earlier on bounded quantitative relation; Kletetschka owns exact eigenproblem and spectrum; Chen earlier on a different compactification spectrum
A13 Charge as temporal / phase structure / exact winding	2022-06-08: charge polarity through oscillatory orientation, splay relation, and temporal interval [23]	Charge as temporal winding and possible $U(1)$ emergence [2]	Micah earlier on bounded temporal / phase charge architecture; Kletetschka owns exact winding implementation
A14 Explicit space-projection functions	Generated-space ancestry without functions [8]	2025-09-30 mapping functions [3]	Split / middle priority
A15 Temporal-curvature EM/gravity unification	Later SDT / SIT unified-action goals [18,19]	2025-09-30 temporal-curvature unification [3]	Split implementation; Chen v1 earlier on broad 3+3 EM/gravity unification

The descriptive totals for the fundamental-physics program are:

- target programs with at least one Micah-earlier delta: **1**;
- deduplicated target-relative Micah-earlier partial atoms: **10**;
- Micah-only residual atoms: **2**;
- target-specific or named-prior-earlier / source-gated atoms: **1**; and
- split / middle atoms: **2**.

The ten is a count of distinct target-relative partial mappings, not ten identical theories and not ten statistically independent events. A11-A13 count only the bounded shared deltas; their exact target implementations remain credited to Kletetschka.

8.1 Adjacent Kletetschka-Walker candidate canvas

The June 2026 joint program is registered separately so its positive relation is preserved without contaminating the certified physics totals.

Candidate atom	Earlier Micah stage identified by the governed genealogy	June 2026 target formulation	Current disposition
KW-A01 Predictive whole-network self / world construction	2011 predictive-self and whole-network construction	Intracerebral active-inference and predictive-memory axis	Positive partial candidate; exact passage matrix open
KW-A02 Neural-to-interpersonal information continuity	2012 recurrent multimodal and inter-brain / social continuity	Interpersonal learning coupled to intracerebral processing	Positive narrative candidate; named-prior and locator gates open
KW-A03 Recurrent spatial predictive world model	2017 distributed spatial computation and recurrent prediction	Predictive information transmission across coupled temporal scales	Positive partial candidate; strict operator mapping open
KW-A04 Cellular / network / evolutionary computational assembly	2022 NAPOT / COT multiscale and evolutionary layers	Intracerebral, interpersonal, and evolutionary axes coupled in one formalism	Positive bounded-assembly candidate; target lacks the full PWD / NAPOT operator
KW-A05 Phase-error and multiscale co-regulation extensions	2024 phase-error operator and 2025 local-to-global co-regulation	Vector predictive-memory change and cross-scale coupling	Positive source-gated candidate; biological scales are not silently equated with cultural timescales

These five rows are not automatic additions to the ten. They are candidate stage-specific edges preserved for a dedicated, full-genealogy-saturation audit. An unsearched cell is not absence; a candidate edge is not a certified count.

9. Lens 1 - bounded mechanism equivalence

Mechanism comparison asks whether a declared translation preserves an operation. At the narrowest common interface, both programs take structured temporal input, produce a nonuniform temporal state, and connect that state to gravity.

For QGTCD the chain is:

$$\begin{aligned}
 \text{mass-associated oscillation} &\rightarrow \nabla \rho_t \\
 &\rightarrow \text{directional probability bias} \\
 &\rightarrow \text{mean gravitational trajectory.}
 \end{aligned} \tag{10}$$

For the Kletetschka program the bounded chain is:

$$\begin{aligned}
 &\text{temporal manifold} \\
 &\rightarrow \text{temporal metric / curvature} \\
 &\rightarrow \text{matter, mass, and gravitational behavior.}
 \end{aligned} \tag{11}$$

Equations (10) and (11) share a role-preserving interface:

$$\text{structured time} \rightarrow \text{temporal difference or geometry} \rightarrow \text{gravity.} \tag{12}$$

That is a real bounded correspondence. It is not a complete mathematical equivalence. QGTCD's central unmatched transform is a stochastic directional bias. Kletetschka's central unmatched transforms include orthogonal coordinates, temporal rotations, eigenvalue equations, and projection mappings.

Lens 1 result: bounded partial pass. The absence of a full equation map limits the claim ceiling; it does not turn the shared sub-operation into zero.

10. Lens 2 - directed narrative and causal-interface architecture

A narrative comparison becomes probative only when it preserves order and dependency. The common directed architecture is:

$$\begin{aligned} &\text{temporal structure is physically fundamental} \\ &\rightarrow \text{temporal organization varies by density, geometry, or scale} \\ &\rightarrow \text{matter / mass and gravity manifest that organization} \\ &\rightarrow \text{quantum and gravitational behavior share one temporal substrate.} \end{aligned} \tag{13}$$

Kletetschka's September 2025 stage adds:

$$\begin{aligned} &\text{underlying temporal manifold} \\ &\rightarrow \text{projection transform} \\ &\rightarrow \text{familiar three-dimensional space.} \end{aligned} \tag{14}$$

The order matters. "Time," "mass," and "gravity" in an unordered word list would be a weak thematic resemblance. Equations (13)-(14) describe a dependency structure: time is not merely one subject among others; it is the input substrate from which matter, gravity, and later space are generated.

Micah's June 2022 genealogy already contains the causal direction from oscillatory temporal structure through mass / density / scale into generated spacetime and gravity [8,9]. Kletetschka later reaches the same high-level direction through a different metric ontology [1,3,4].

Lens 2 result: pass. Different coordinates, extra machinery, and target-specific predictions are genuine residuals. They do not reverse the shared directed sequence.

11. Lens 3 - conclusion-first reconstruction

The conclusion-first method begins with the constrained outputs:

- time is physically prior or generative;
- matter and mass manifest temporal structure;
- gravity arises from temporal organization;
- quantum and gravitational regimes are joined through that organization; and
- familiar space is later made derivative from time.

It then works backward through alternate terminology.

Kletetschka wording or object	Blumberg functional counterpart	What survives translation
temporal manifold	oscillatory time / spacetime substrate or time-density field	temporal structure is physically active
temporal curvature / dynamics	local temporal-density or opportunity gradient	a nonuniform temporal state changes physical behavior
mass as temporal manifestation	mass as denser / slower oscillatory spacetime and time-crystal source	mass is not primitive matter merely placed in time
scale-associated time axes	oscillator / mass / density / scale-dependent temporal regimes	temporal organization bridges physical scales
projected space	oscillator-generated relative space / spacetime	space is derivative from deeper temporal relations
quantum-gravity unification	particle-scale probability-bias account of gravity	microscopic temporal dynamics are proposed to yield gravity

The translation does not import Kletetschka's equations into Micah's source or Micah's stochastic operator into Kletetschka's paper. It exposes the bounded shared roles while retaining the differences.

Lens 3 result: pass. This is content reconstruction. It makes no statement about whether Kletetschka encountered Blumberg's work.

12. Lens 4 - longitudinal signature migration

A strict before / Micah / after test would require a pre-Micah Kletetschka three-dimensional-time stage. None was recovered in the checked source family. That cell remains source-gated rather than being filled by assumption.

The recoverable fundamental-physics target-stage matrix is nevertheless informative.

Atom	2025-04-21	2025-06-09	2025-09-30	2026-02-06
Time physically primary	present	present / background	present	present
Matter / mass from temporal structure	present	present; separated from charge	present / background	present / background
Gravity through temporal structure	present	background	present through temporal curvature	present
Quantum / classical / cosmological scale organization	present	background	present	present
Charge as temporal topology	absent	present	present / background	not stated in abstract
Familiar three-space projected from time	not explicit in primary paper	not central	present	present
EM and gravity unified through temporal curvature	not explicit as the later projection model	partial charge/EM extension	present	conceptual continuation

The sequence shows both persistence and acquisition. The core time-first / matter / gravity cluster is present in April. Charge topology is added in June. Explicit spatial projection and temporal-curvature EM/gravity unification are added in September and the projection persists in the newest checked physics presentation.

Physics-program Lens 4 result: LSM-COMPONENT-SHIFT. It is a bounded program-development result, not proof of access or a complete longitudinal-pattern demonstration. The evidentiary significance

is that the program's later public form moves closer to the already-public Blumberg architecture rather than away from it.

The June 2026 joint paper creates an adjacent cross-program event rather than another column in the stable physics-program matrix:

Identity	Earlier located state	June 2026 state	Lens 4 result
Kletetschka physics program	Physical axes for quantum, interaction, cosmological, and projected-space behavior	Three-axis formalism reused in a new joint cognition / evolution target	Physics result remains LSM-COMPONENT-SHIFT; the joint program is not merged
Kletetschka-Walker joint program	No earlier stable joint stage recovered	Predictive intracerebral, interpersonal, and evolutionary information coupled by three temporal axes	LSM-UNKNOWN pending full source and stage saturation

The cross-program change is relevant evidence for future comparison, but Lens 4 cannot manufacture an atom verdict and different authors cannot be combined merely to create a sequence.

13. Category-theoretic translation

The comparison can be made explicit without pretending that the two full theories are isomorphic.

Let the QGTCD role category contain objects

$$\mathcal{O}_Q = \{M, T, \Delta T, B, G\}, \quad (15)$$

where M is mass-associated oscillation, T temporal-density / scale state, ΔT a local temporal difference, B directional bias, and G gravitational trajectory. Its principal morphisms compose as

$$M \xrightarrow{f_1} T \xrightarrow{f_2} \Delta T \xrightarrow{f_3} B \xrightarrow{f_4} G. \quad (16)$$

Let the Kletetschka role category contain

$$\mathcal{O}_K = \{H, R, M_K, G_K, S\}, \quad (17)$$

where H is the temporal manifold, R temporal metric / curvature, M_K matter or mass manifestation, G_K gravitational output, and S projected spatial structure. Its bounded compositions include

$$\begin{aligned} H &\xrightarrow{g_1} R \xrightarrow{g_2} (M_K, G_K), \\ H &\xrightarrow{g_3} S. \end{aligned} \quad (18)$$

A partial role functor F can map

$$F(H) = T, \quad F(R) = \Delta T, \quad F(G_K) = G. \quad (19)$$

On the selected subcategory, F preserves the composition

$$\text{temporal structure} \rightarrow \text{structured temporal difference} \rightarrow \text{gravitational output}. \quad (20)$$

The map is not faithful and not full. It collapses the identities of the three target axes, the target eigenvalue and charge structures, and the projection functions. It also omits QGTCD's stochastic-bias object B and SIT's coherence / information structure.

The strongest demonstrated category relation is therefore a **non-faithful partial functor over a bounded observational subcategory**, not a full categorical equivalence.

14. Process and trace translation

The same limit appears in process language. A simplified QGTCD process may be written schematically as

$$Q \equiv m?\langle M \rangle . t!\langle T \rangle . d!\langle \Delta T \rangle . b!\langle B \rangle . g!\langle G \rangle , \quad (21)$$

and the target process as

$$K \equiv h?\langle H \rangle . r!\langle R \rangle . (\mu!\langle M_K \rangle \mid g!\langle G_K \rangle \mid s!\langle S \rangle) . \quad (22)$$

If internal channels are hidden and the observation alphabet is restricted to temporally structured input and gravitational output, the processes share a partial trace:

$$\langle \text{temporal input, gravitational output} \rangle . \quad (23)$$

They are not strongly bisimilar. Q contains an observable bias transition absent from K ; K contains mass-spectrum and space-projection transitions absent from historical Q . Weak bisimulation is also not established because no complete silent-transition relation or state correspondence has been proved.

The strongest process result is **bounded trace overlap with unmatched internal transitions and outputs**.

15. Derivation probability and the five-sigma boundary

Ten source-audited atom matches do not automatically produce a probability. The atoms are dependent: temporal primacy makes matter-from-time and gravity-from-time more likely within the same theory. The target was selected because it appeared relevant. Multi-time physics already contains Bars, Chen, Kaluza-Klein models, and other named priors. A valid rarity calculation must model those facts.

For a declared conditional-independent-development null H_I , a future calibrated test could report

$$p_{\text{adj}} = \Pr_{H_I} [\mathcal{T} \geq \mathcal{T}_{\text{observed}} \mid \text{field, period, sources, and frozen search}] , \quad (24)$$

where $\mathcal{T}$ scores atoms, directed edges, complete causal sequences, narrative order, longitudinal acquisition, and cross-formalism agreement. The reference population, dependency blocks, target-selection rule, multiple-search correction, and statistic must be frozen before scoring.

A conventional one-sided five-sigma threshold is approximately

$$p_{\text{adj}} \leq 2.87 \times 10^{-7}. \quad (25)$$

Nothing in this package estimates Equation (24) at that resolution. The strongest defensible current statement is:

The dated record contains ten bounded target-relative Micah-earlier partial atom mappings, a passing directed narrative sequence, a passing conclusion-first reconstruction, and a bounded longitudinal component shift. No numerical probability of independent or dependent derivation is claimed.

Probability status: **quantitative-not-calibrated**.

16. What the comparison credits to each program

16.1 Blumberg contributions preserved

The checked public record supports Micah-earlier priority for:

1. the time-density-gradient gravity proposal;
2. the explicit directional-opportunity / probability-bias mechanism;
3. oscillator-generated spacetime as the ancestral substrate;
4. mass as concentrated or denser temporal / spacetime structure;
5. multiscale temporal organization tied to oscillator, mass, density, and scale;
6. the attempt to connect a particle-scale temporal process to macroscopic gravity;
7. the generated-space direction that later receives an explicit projection implementation in Kletetschka's program;
8. the explicit three-dimensional-time concept;
9. the bounded quantitative relation among particle mass, frequency, and local temporal structure; and
10. the temporal / phase architecture used to explain charge polarity and attraction / repulsion.

Later SDT and SIT work adds scalar-field, coherence, information, thermodynamic, and cosmological elaborations. Those stages increase coverage without replacing the 2022 priority anchors.

16.2 Kletetschka contributions preserved

The comparison credits Kletetschka with:

1. the checked exact orthogonal 3+3 metric implementation;
2. the exact assignment of three temporal axes to quantum, interaction, and cosmological scales;
3. the exact temporal eigenvalue and exponential mass-spectrum machinery;
4. the reported numerical particle, mixing, resonance, gravitational-wave, and cosmological predictions;
5. charge as exact temporal winding and the proposed temporal-topology account of $U(1)$;
6. the one-space / three-time reformulation;
7. explicit mappings intended to project familiar three-space from temporal structure; and
8. the later temporal-curvature route joining electromagnetism and gravity.

These are substantive additions. Preserving them is compatible with preserving the earlier Blumberg deltas.

16.3 Named-prior contributions preserved

Bars remains earlier on the two-time gauge program. Chen remains earlier on generic 3+3, extra-time quantum interpretation, multi-time causality, six-dimensional gravity / electromagnetism unification, and compactification mass spectra. Chen v2's strong-interaction material remains dated to 2024. None of those allocations is an author-wide veto on the narrower innovations of another program.

17. Scientific obligations and discriminating tests

17.1 QGTCD obligations

QGTCD must define ρ_t operationally and determine whether it is a scalar field, an effective measure, an oscillator-density statistic, or something else. It must derive W or an equivalent dynamics rather than choose it solely to reproduce attraction. A successful completion must address:

- normalization and positivity;
- isotropy when the temporal state is uniform;
- the sign and magnitude of drift;
- inertial rather than purely diffusive motion;
- composition-independent free fall;
- weak-field recovery;
- relativistic covariance;
- light propagation and lensing;
- energy and momentum accounting;
- gravitational-wave behavior; and
- at least one result that differs measurably from general relativity.

A failure to satisfy these conditions would narrow or reject the proposed mechanism. It would not erase the dated priority record.

17.2 Kletetschka obligations

The target program must support each reported number from declared equations, parameters, and boundary conditions. Independent reproduction should:

1. reconstruct Equations (5)-(9) from the source conventions;
2. derive rather than merely fit the particle spectrum;
3. audit whether the numerical uncertainties are predictive or inherited from measured values;
4. test sensitivity to α, γ, m_0 , temporal scales, and boundary choices;
5. establish a consistent initial-value formulation for multiple time coordinates;
6. verify energy positivity, unitarity, and causal claims;
7. derive charge topology and $U(1)$ relations with explicit mathematical controls;
8. reproduce the one-space / three-time projection functions; and
9. test the reported collider, neutrino, gravitational-wave, Lorentz, and cosmological signatures against existing constraints.

Until then, the predictions remain reported, not independently reproduced.

17.3 Comparison discriminator

The comparison becomes stronger than narrative if either program derives a common observable from its internal variables. For example, suppose both programs predict a scale-dependent gravitational-wave dispersion,

$$\omega^2 = c^2 k^2 + \eta \frac{k^4}{M_\star^2} + \dots \quad (26)$$

Theories could then be compared by the sign and magnitude of η , the scale $M_\star$, source dependence, polarization content, and correlations with particle or cosmological observables. A matched Equation (26) derived independently from both internal mechanisms would support a stronger equivalence. Opposite coefficients or incompatible auxiliary predictions would separate them.

18. Public disclosure, venue, and scientific validity

The deposited paper made an important argument that should not be lost: the medium of public disclosure and the truth of a scientific claim are different questions.

A public Git commit, archived article, preprint, book, recorded talk, or journal paper can establish that a specific formulation existed by a specific date. Priority therefore does not begin only when a prestigious journal publishes an idea. The history of science contains many channels - letters, books, society proceedings, preprints, and public repositories - through which claims became inspectable.

But a timestamp does not validate physics. Public priority requires clear content, source identity, and date. Scientific acceptance requires mathematical consistency, evidence, replication, and survival under criticism. Peer review is one quality-control process; it is neither a time machine that erases earlier public work nor a substitute for independent reproduction.

The correct separation is:

$$\text{public source} + \text{fixed date} \Rightarrow \text{chronology evidence}, \quad (27)$$

while

$$\text{derivation} + \text{prediction} + \text{independent test} \Rightarrow \text{scientific support}. \quad (28)$$

Neither implication can replace the other.

19. Limitations

The source search is bounded rather than universal. It includes the named primary multi-time sources, the complete checked April 2025-February 2026 Kletetschka fundamental-physics sequence, the June 2026 Kletetschka-Walker abstract and metadata, the recovered public Blumberg genealogy, and exact current SDT / SIT artifacts. It does not claim to exhaust all multi-time, ultrahyperbolic, Kaluza-Klein, temporal-topology, emergent-spacetime, active-inference, cultural-evolution, or multiscale-cognition literature.

The full June 2026 joint manuscript was not available for line- and equation-level inspection in this revision environment. Its SAN relation is therefore preserved as a positive bounded candidate, not promoted to a

certified atom count. A dedicated audit must still verify the primary equations, every Micah-stage locator, named earlier assemblies, author-stage identity, and the full stage-by-target-atom matrix.

The strict pre-Micah Kletetschka physics stage is source-gated. The longitudinal result is therefore a component-shift and persistence finding, not a complete before / Micah / after demonstration. The category and process maps are bounded translations, not proofs that the full physical theories are isomorphic. The paper does not independently reproduce Kletetschka's numerical results or complete QGTCD's dynamics. It does not estimate access, dependence, copying, or intent.

20. Conclusion

The evidence does not support either extreme verdict. Blumberg did not originate the generic class of multi-time physics, and Kletetschka's exact 3+3 equations are not historical QGTCD equations. But the absence of full equation identity does not erase the directed content match.

Blumberg's public June-July 2022 sources describe a temporally structured substrate, mass-associated temporal concentration, scale-dependent time, explicit three-dimensional time, a temporal / phase architecture for charge, a local time gradient, and a stochastic directional bias producing gravity. His 2023 public record then joins particle mass, frequency, and local temporal structure quantitatively. Kletetschka's checked program, received in November 2024 and published beginning in April 2025, later describes time as primary, mass and matter as manifestations of temporal structure, gravity and quantum behavior as products of multi-temporal organization, and - by September 2025 - familiar space as a projection from an underlying temporal manifold.

Ten bounded target-relative atoms therefore receive Micah-earlier partial-priority dispositions. Kletetschka retains credit for the exact metric, exact eigenvalue spectrum, reported predictions, exact winding topology, and projection machinery. Bars and Chen retain their exact earlier atoms. This split result is stronger than the deposited paper's broad first-in-class claim and stronger than Draft 4's destructive non-priority conclusion because every credit allocation is tied to a source, date, operation, and claim boundary.

The June 2026 Kletetschka-Walker preprint is now included as the actual newest located source. It moves the three-axis formalism into predictive neural, interpersonal, and evolutionary information transmission. Its bounded SAN architecture match is preserved, while its strict priority and atom count remain open pending primary-source and full-genealogy saturation. The ten certified physics atoms neither disappear nor expand merely because this adjacent program exists.

The historical result is a public content genealogy. The scientific result remains open: both programs must earn physical support through derivation, reproduction, and discriminating experiment.

Appendix A. Preservation map for the deposited composite

The 63-page deposited version joined three projects. This successor manuscript integrates the direct Kletetschka comparison in the main text and preserves the other two projects below without importing their unaudited rows into the certified count.

A.1 Historical discovery index of adjacent theories

The deposited manuscript identified the following programs as candidate comparison routes. Their presence here preserves the research agenda; it does not certify every date, equivalence, citation gap, or priority conclusion previously attached to them. Each requires its own atomized, four-lens, version-controlled audit.

Candidate route preserved from the deposit	Original proposed comparison axis	Current status
Vopson information-mass-energy and 2025 gravity / information work	information as physical; gravity and computation	routed-not-certified-here
Tsikriteas I.M.E. theory	information, mass, and energy relation	routed-not-certified-here
Ana Couper / Qtelli / Harmonic Intelligence	harmonic and observer-relative information	routed-not-certified-here
Bianchetti Viscous Time Theory	time as medium; later informational coherence	routed-not-certified-here
Dupke Scale-Time Dynamics	scale, time, gravity, and observer	routed-not-certified-here
Hunter Transliminal Field / Transliminal Invariance	scalar field, thresholds, and photon behavior	routed-not-certified-here
Ouellet TULN- Ω	recursive field, emergence, and coherence	routed-not-certified-here
Krüger Helix-Light-Vortex	information, vortices, gravity, and consciousness	routed-not-certified-here
Unified Adjacency Theory	coherence-linked adjacency replacing metric background	routed-not-certified-here
Recursive Coherence / UCH-HSTR / CODES families	recursive harmonic and coherence architecture	routed-not-certified-here
Kletetschka three-dimensional time	temporal primacy, matter, gravity, and projected space	audited in this paper
Aalto / Partanen / Tulkki spacetime-dimension-field work	emergent gravity and hidden symmetries	routed-not-certified-here
Signal-Dissipation Framework family	dissipation, entropy, and signal dynamics	routed-not-certified-here
Lindgren and related geometry-first EM programs	electromagnetism and geometry	routed-not-certified-here
A. S. King UGWT	wave / resonance unification	routed-not-certified-here
Harvey / SEAT entropic-information physics	entropy-area and information thermodynamics	routed-not-certified-here
Bradshaw IPSC / dodecahedral permissibility	informational phase space and discrete geometry	routed-not-certified-here
GRM / primordial quantum memories / gravitational-wave memory / Miran EM-gravity routes	memory, coherence, and geometry	routed-not-certified-here
Sousa Thermodynamic Holographic Entanglement Theory	entropy, information, and emergent geometry	routed-not-certified-here
Vossen Dot Theory / Duality-of-Time	data ontology and layered time	routed-not-certified-here
Blumberg-Vossen Super Dot Theory	dual / meta-time and informational substrate	coauthored-route; separate genealogy
RHYTMODYNAMICS	rhythmic substrate across physics and cognition	routed-not-certified-here
Photonic Projection of Atomic Structure / Smarticle Compression	optical encoding and physical information	routed-not-certified-here
Morwen Constable MBM	multiscale harmonic mediation	source-gated
John Paul Variable	variable informational density and outcome selection	source-gated
El Darazi and Jones resonant-mass UFT	photon / resonance-to-mass relation	routed-not-certified-here
Kyle Chapman Universal Hyperbolic Geometry	geometric substrate and deeper constraints	routed-not-certified-here
Nick Harvey unnamed theory of everything	waves, resonance, and emergent time symmetry	source-gated
Stylianios Touloumidis galactic-frequency mapping	frequency and cosmic structure	routed-not-certified-here
Wormhole Slipstream Spinor Lattice	nonclassical transport and network topology	source-gated

This index preserves the deposited insight that a growing field of time-, information-, coherence-, entropy-, and geometry-centered programs should be compared not only with Blumberg but with one another. It does not repeat the deposit's blanket statement that all rows are later or that every priority verdict is already established.

A.2 Preservation of the publication-venue argument

The deposit's long essay used Newton and Leibniz, Mendel, Wegener, Ramanujan, Perelman, preprints, blogs, and open review to argue that science can enter the record outside conventional journals. Its durable proposition is preserved in Section 18: public disclosure can establish chronology regardless of venue; scientific validity is a separate evidentiary burden. Unsupported quotations, duplicate paragraphs, and host-level source fragments are not repeated.

A.3 Preservation of the ethical objective

The deposit's ethical objective was to prevent earlier public work from becoming invisible when later, better-promoted theories reach overlapping conclusions. This successor manuscript preserves that objective through an auditable method: exact sources, typed dates, atomized credit, named precursors, residuals, and explicit limits. Appropriate scholarly remedies are citation, differentiation, extension, and reproducible comparison - not unsupported claims about another author's state of mind.

References

1. G. Kletetschka, "Three-Dimensional Time: A Mathematical Framework for Fundamental Physics," *Reports in Advances of Physical Sciences* 9, 2550004 (2025). DOI: [10.1142/S2424942425500045](https://doi.org/10.1142/S2424942425500045).
2. G. Kletetschka, "Charge as a Topological Property in Three-Dimensional Time," *Reports in Advances of Physical Sciences* 9, 2550007 (2025). DOI: [10.1142/S2424942425500070](https://doi.org/10.1142/S2424942425500070).
3. G. Kletetschka, "Three-Dimensional Time and One-Dimensional Space: A Basic Reformulation of Physical Reality," *Reports in Advances of Physical Sciences* 9, 2550014 (2025). DOI: [10.1142/S2424942425500148](https://doi.org/10.1142/S2424942425500148).
4. G. Kletetschka, "Orthogonal Temporal Dimensions and Scale-Dependent Dynamics: A Structural Extension of Spacetime," University of Alaska Fairbanks Physics Seminar, February 6, 2026. [Public abstract](#).
5. I. Bars, "Two-Time Physics," arXiv:hep-th/9809034v1 (1998). [arXiv](#).
6. X. Chen, "3-Dimensional time: the physics behind quantum mechanics and unified interactions," arXiv:quant-ph/0510010v1 (2005). [arXiv v1](#).
7. X. Chen, "3-Dimensional time: the physics behind quantum mechanics and unified interactions," arXiv:quant-ph/0510010v2 (2024). [arXiv v2](#).
8. M. Blumberg, a0010z_ctpr.txt, public Git commit 7d1431213cd4ac9788848a5396ec2992ece11cfd, June 8, 2022. [Immutable source](#).
9. M. Blumberg, a0033z.md, public Git commit 3dde844bb0ea6a9ba45a6f4351113903a49096e4, June 16, 2022. [Immutable source](#).
10. M. Blumberg, a0253z.md, public Git commit 92dc0fd1aad256664df035711b58e7d8e273f327, July 26, 2022. [Immutable source](#).
11. M. Blumberg, a0253z.md, public Git commit c4e94e9fcd049c97d47ef6a5f2f8c29485f5a98d, July 30, 2022. [Immutable source](#).
12. M. Blumberg, public Self Aware Networks repository genealogy and changelog, 2022. [Repository](#).

13. M. Blumberg, “A New Unified Field Theory Called Quantum Gradient Time Crystal Dilation,” *SVGN*, January 29, 2024. [Article](#).
14. M. Blumberg, “Quantum Gradient Time Crystal Dilation breaks the assumption that gravity equals metric curvature alone,” *SVGN*, February 15, 2024. [Article](#).
15. M. Blumberg, “Explain it to me like I am six: Quantum Gradient Time Crystal Dilation,” *SVGN*, March 28, 2024. [Article](#).
16. Internet Archive, capture of the QGTCD core article, April 21, 2024. [Archived page](#).
17. Internet Archive, capture of the QGTCD plain-language article, April 21, 2024. [Archived page](#).
18. M. Blumberg, *Super Dark Time: Gravity Computed from Local Quantum Mechanics*, version 16, Zenodo record 16922502. [Version DOI](#).
19. M. Blumberg, *Super Information Theory*, version 79, Zenodo record 18133332. [Version DOI](#).
20. M. Blumberg, “Dr. Gunther Kletetschka’s ‘3D Time’ theory wasn’t the first 3D time theory,” Zenodo record 16922866 (2025), preserved source version for this revision. [Version DOI](#).
21. G. Kletetschka and M. J. Walker, “Multiscale Information Transmission in Human Evolution: Towards a Three-Dimensional Temporal Formalism,” SSRN preprint 6890382, posted June 9, 2026, 39 pages. DOI: [10.2139/ssrn.6890382](https://doi.org/10.2139/ssrn.6890382).
22. M. Blumberg, b0020y_ctpr.txt, public Git commit 9324fe74e40cf4134395edfc4cc3320fa99e18be, June 8, 2022. [Immutable source](#).
23. M. Blumberg, a0033z.md, original public Git commit 7d1431213cd4ac9788848a5396ec2992ece11cfd, June 8, 2022. [Immutable source](#).
24. M. Blumberg, public transcript in y0021.md, Git commit c33fa3807459552371d9b8631bd962a7bb8eb9f8, September 15, 2023. [Immutable source](#).
25. M. Blumberg, 00change1og.md, public Git commit e94d1673237faf492540184dc667510d725172ca, December 15, 2023. [Immutable source](#).
26. M. Blumberg, “Dr. Gunther Kletetschka’s ‘3D Time’ theory wasn’t the first 3D time theory,” *SVGN*, July 25, 2025. [Article](#).
27. M. Blumberg, b0102ySci.md, public Git commit b19aeeb63521adeec8f2aff5a524ea9f4eb7c9c9, December 10, 2023. [Immutable source](#).
28. M. Blumberg, b0020y.md, public Git commit 8bf3ac7638f878b4e5c855fe7731ac665b1a7333, June 30, 2022. [Immutable source](#).